

Sourav Das Partha

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RESEARCH PROFILE

Urban and transport planner with a Master in GIS for Environment and Development and over seven years of applied experience in travel demand modelling, spatial analysis, and large-scale transport survey data across Bangladesh. Experienced across the full four step travel demand modelling pipeline, from survey design and data collection to trip generation, trip distribution, mode choice, and trip assignment, including origin destination matrix development and volume estimation within a comprehensive modeling framework. Skilled in GIS based spatial and network analysis, network design and optimization, and statistical analysis using R, Python, and Stata. Work spans mass transit planning, transit-oriented development, land use and real estate demand studies, and road safety research, with a focus on data driven, policy relevant urban systems research.

TECHNICAL SKILLS

Statistical analysis & programming: R, Python and Stata

GIS & spatial analysis: ArcGIS Pro, ArcGIS Desktop, QGIS

Transport modelling: TransCAD, PTV Vissim, SIDRA Intersection

Design & CAD: AutoCAD 2D, Civil 3D, SketchUp, Adobe Illustrator

Productivity: Microsoft Word, Excel, PowerPoint, Project

EDUCATION

MS in GIS for Environment and Development

Summer 2019 to Summer 2022

Department of Geography and Environment, Jahangirnagar University, Dhaka | CGPA 3.58 / 4.00

Bachelor of Urban and Regional Planning

January 2014 to December 2017

Department of Urban and Regional Planning, Jahangirnagar University, Dhaka | CGPA 3.04 / 4.00

RESEARCH EXPERIENCE

Research in Progress

- Travel Demand Modelling with Limited Data: Adapting the Four-Step Framework for Low-Resource Urban Contexts. (Ongoing)
- Microscopic Modelling of Traffic Emissions in Heterogeneous Traffic Conditions. (Ongoing)
- The Role of Electric Three-Wheelers in Sustainable Urban Mobility Transitions. (Ongoing)
- Spatial Analysis of Road Crash Patterns and Severity in Gazipur City Corporation. (Ongoing)

Theses

- Traffic Flow and Bus Demand Rationalization: A Case Study of the Dhanmondi 27 to Lalbagh Corridor (MS Thesis).
- Perception of Tourists about Transportation in Selecting Tourist Spots: A Case Study of Tanguar Haor and Jadukata River (Undergraduate Thesis).

PROFESSIONAL EXPERIENCE

Junior Traffic and Transportation Planner, Nippon Koei Bangladesh Ltd. March 2024 to Present

Comprehensive Transport Master Plan with Pre-Feasibility Study of Mass Transit Network and Conceptual Design of Truck Terminal, Gazipur City Corporation (funded by DTCA)

- Conducted freight and truck surveys and collected terminal demand data, handling large and complex field datasets to support freight demand estimation for the city-scale transport network.
- Developed a four-step travel demand model in TransCAD from scratch, including the public transport component, and produced final outputs comparing existing and forecast demand.

- Applied a log-linear (multiplicative) production model for home-based trips and ran a nested logit mode choice model within the four-step framework.
- Carried out network preparation, OD matrix development, and highway and public-transport network design, with supporting spatial and network analysis in GIS.
- Examined the interlinkage between transport and urban planning, developing sustainable transport options, road network and demand management solutions, and intermodal integration.
- Identified effective policies to close the land use–transport integration gap, and prepared technical reports to support planning and modelling decisions.
- Applied technical skills in GIS software (ArcGIS and QGIS) and spreadsheet-based data analysis, working within a multidisciplinary team.

Data Collection Survey on Public Transport Mode and Operation in Dhaka (funded by JICA)

- Conducted public and private bus operations surveys; reviewed regulations, ownership, registration, maintenance and parking arrangements.
- Estimated fleet size, identified restricted zones, analyzed operational associations, and conducted driver interviews; provided recommendations and reporting.

Policy and Guidelines for Transit-Oriented Development along Mass Transit Corridors (funded by JICA)

- Supported land-use and real-estate demand studies, compiling location characteristics and economic/infrastructure conditions from real-estate valuation surveys.
- Developed GIS/CAD databases and maps; mapped existing land use, buildings, road networks, technical infrastructure and land boundaries.

Junior Urban / Transport Planner, Engineering and Planning Consultants Ltd. September 2021 to February 2024

Traffic Study cum Junction Design, Dhaka City Neighborhood Upgrading Project, DSCC (funded by World Bank)

- Supervised traffic and transportation surveys and assisted in preparing Traffic Management Plans (TMP).
- Contributed to intersection design in Civil 3D, producing intersection layouts, land-use maps and existing traffic-flow maps.

Western Economic Corridor & Regional Enhancement Program (WeCARE) Phase 1 (funded by World Bank)

- Supervised traffic surveys and prepared thematic and road-connectivity maps from field and survey data.
- Generated land-use maps and Digital Elevation Model (DEM) outputs to support flood-risk and terrain assessment.

Feasibility Study for Improvement of Intersections Including Signals, DSCC (funded by DSCC)

- Analyzed traffic, pedestrian, public-transport, OD, travel-time, delay and parking survey data, including LOS and ADT calculation and OD matrix preparation.
- Contributed to intersection design in Civil 3D and prepared roadway layouts and analytical maps.

Junior Urban Planner, Image Research and Consultancy Ltd. October 2018 to August 2021

Slum-based Citizen Action Network (SCAN) Project (funded by Solidarités International)

- Conducted GPS-based boundary demarcation and collected ground control points for satellite image rectification.
- Built point and polygon shapefiles integrating digitization with field verification; prepared base maps and final GIS databases.

Urban LEDS II: Open Green Space Plan, Rajshahi City Corporation (funded by UN-Habitat)

- Conducted urban natural-asset and tree surveys (GPS-based tree mapping and species identification) and compiled a tree handbook.
- Produced natural-asset and ecosystem maps from rectified satellite imagery and field verification.

INTERNSHIPS & VOLUNTEERING

- Project Volunteer, Bridging the Gap Project, Save the Children Bangladesh (June to December 2017).
- Volunteer, Cities Forum: Building Knowledge Networks and Partnerships for Sustainable Urban Development in Bangladesh (February to April 2020).

TRAINING & CERTIFICATIONS

- Travel Demand Modelling, Dhaka Transport Coordination Authority (DTCA), Dhaka (20 May to 8 June 2026).
- Urban Transportation Planning, Bangladesh Institute of Management (BIM), Dhaka (14 to 19 January 2025).
- Introduction to R for Planning and Social Science, Bangladesh Institute of Planners (BIP), Dhaka (13 to 14 December 2024).
- Research Methodology, Global Platform Bangladesh, Dhaka (14 to 18 November 2018).

PROFESSIONAL AFFILIATIONS

- Member, Bangladesh Institute of Planners (BIP-2056).
- Board Member (National and International Liaison), 5th Working Committee, JU Planners (Alumni Organization, Dept. of URP, Jahangirnagar University).
- Joint Secretary-1 (Information and Communication), 7th Working Committee, JU Planners.

ENGLISH LANGUAGE PROFICIENCY

IELTS Academic: Overall 7.5 | Listening 8.0 | Reading 7.5 | Speaking 7.0 | Writing 6.5

REFERENCES

1. Dr. Afsana Haque, Associate Professor, Department of URP, Jahangirnagar University • afsana@juniv.edu
2. Md. Aminul Islam, Deputy General Manager, Nippon Koei Bangladesh Ltd. • aminul-is@nkbangladesh.co.bd